1. $\begin{bmatrix} 1 & 3 & 5 \\ -2 & 2 & 6 \\ 3 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 0 & 8 & 16 \\ 0 & -8 & -16 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 0 & 8 & 16 \\ 0 & 0 & 0 \end{bmatrix}$ Zero vector means the vectors are linearly dependent

2. $\left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 1 & 0 & 2 & b_2 \\ -3 & 4 & 6 & b_3 \end{array}\right]$

$= \left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 0 & -2 & 5 & b_2-b_1 \\ 0 & 10 & -3 & b_3-3b_1 \end{array}\right]$

$= \left[\begin{array}{ccc|c} 1 & 2 & -3 & b_1 \\ 0 & -2 & 5 & b_2-b_1 \\ 0 & 0 & 22 & b_3+5b_2-2b_1 \end{array}\right]$

3. $Ax=0$, set $Bx=y$

$ABx = Ay$

Since $Ax=0$, $Ay=0$

So AB is invertible.

4.

$\dim \text{Nul } A^T = 8 - \text{rank } A^T$

to find the smallest null space,

$\text{rank } A^T = \text{rank } A \leq \min(8,5) = 5$

so, $\dim \text{Nul } A^T = 8-5=3$

5. $A = \begin{bmatrix} -14 & 12 \\ -20 & 17 \end{bmatrix}$

$\det(A - \lambda I) = \begin{bmatrix} -14-\lambda & 12 \\ -20 & 17-\lambda \end{bmatrix}$

$= -238 - 17\lambda + 14\lambda + \lambda^2 + 240$

$= 2 - 3\lambda + \lambda^2$

$\cancel{\lambda = \frac{-3 \pm \sqrt{9-8}}{2}}$

$\cancel{= \frac{-3 \pm 1}{2}}$

$\cancel{= -2, -1}$

$= 2 - 2\lambda - \lambda + \lambda^2$

$= 2(1-\lambda) - \lambda(1-\lambda)$

$\lambda = 2, 1$

$\lambda = 1$

$\begin{bmatrix} -15 & 12 \\ -20 & 16 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$

$\lambda = 2$

$\begin{bmatrix} -16 & 12 \\ -20 & 15 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

6. Any sheer will make u & v not orthogonal to each other.

For example: $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$

7. $\frac{\begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}} \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$

$= \frac{2 - 6 + 5}{1 + 9 + 25} \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$

$= \frac{1}{35} \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix}$

$\begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} - \frac{1}{35} \begin{bmatrix} 1 \\ -3 \\ 5 \end{bmatrix} = \begin{bmatrix} 69/35 \\ 73/35 \\ 6/7 \end{bmatrix}$

8. Invertable matrix only have one solution.